

Problem: Security for Robotic Models

- Records of Adversarial Manipulations
- Unexplored Threat Landscape
- No Security Trustworthiness Benchmarks

Challenge 1: Data/Model Adoption

- Heterogeneous Datasets/Tasks/Models
- Cross-embodiment Generalization

Challenge 2: Multimodal Integration

- Vision/Language Perturbations
- Vision-language Coupling/Fusion
- Action-tokens and Physical Impact

Challenge 3: Limited Adversarial Access

- Transferability and Surrogate-based Threats
- Transferability across Models/Embodiments
- Costs and Efficiency of Threats

Challenge 4: Robustness Gaps

- Limited Evaluation Framework
- Action-space Defenses and Hardening
- Limited Defenses/Interpreters/Benchmarks

Ideas: Towards Security

- Vulnerability Analysis in Robotic Models (RMs)
- Vulnerability Analysis under Limited-Access
- Robustness/Safety/Interpretability Benchmarks

T0: Repository for Models and Data

- Multimodal Robotics Datasets/Models
- Cross-embodiment Model Zoo

T1: Full-Access Attacks on RMs

- Action-Bin Boundary Perturbation (ABBP)
- ABBP with Multimodal Variants
- Evaluation Grounded in Action/Physical Space

T2: Limited-Access Attacks on RMs

- Surrogate-Transfer Action Perturbation (Gray-box)
- Temporal Trajectory Drift Attack (Black-box)
- Extending to Evasion and Privacy Attacks

T3: Defense, Safety, and Interpretation

- Temporal Action Consistency Detector
- Action-Grounded Adversarial Fine-Tuning
- Defenses/Safety/Interpreters Benchmarks

